605.486 – Android Development

Module 23
Google Cloud Messaging

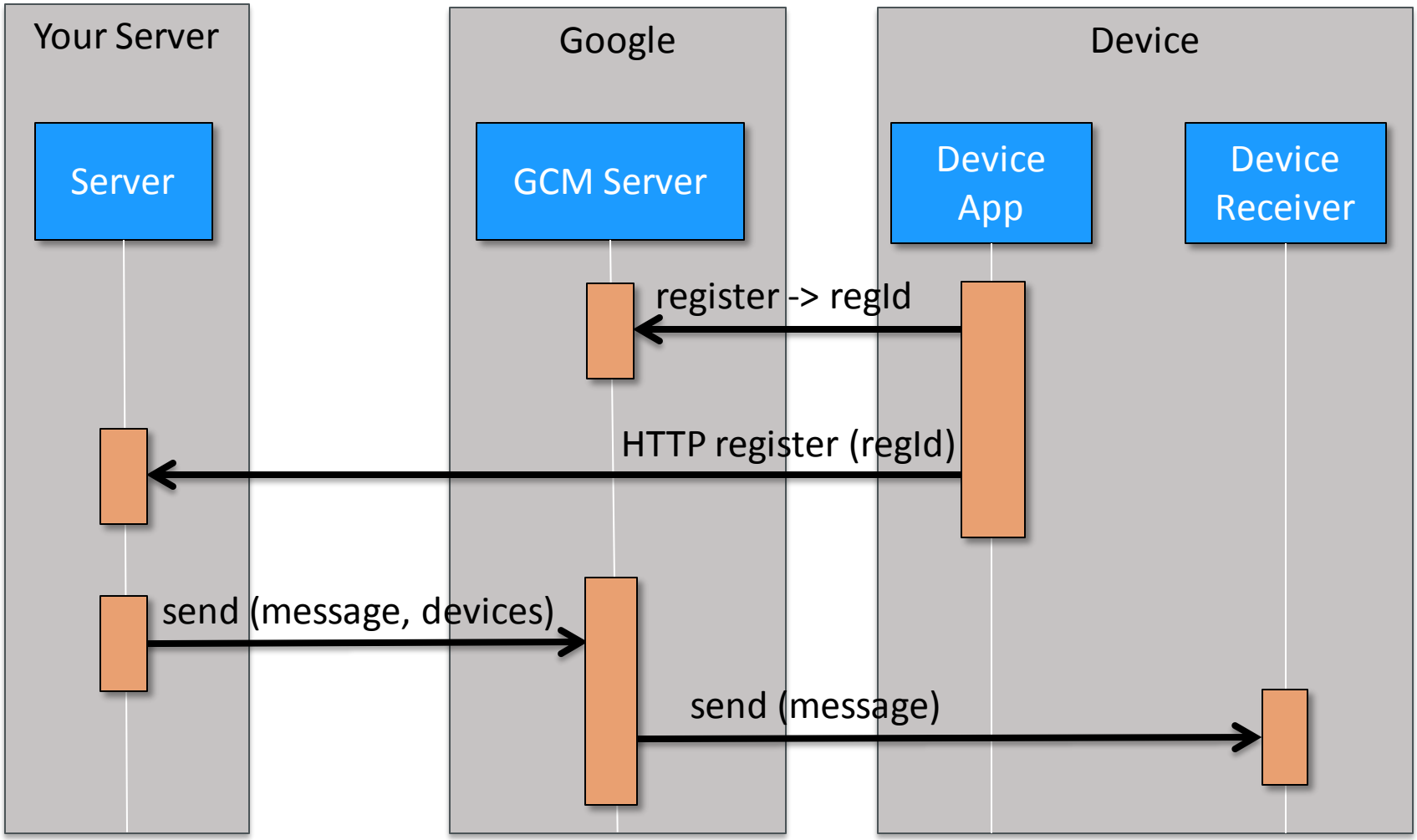
Concepts

- Allows *push* from server to client devices
- Only communicate with clients when needed
- Communication held if device offline
- Reduce bandwidth, battery, server
- FREE!

Google Cloud Messaging

- http://developer.android.com/google/gcm
- Your device registers with GCM server
- Your server pushes message to GCM
- GCM pushes message to registered devices
  - Holds up to 100 msgs, 4k data each, 4 weeks
  - Allows 1000 devices per request
  - Can collapse data (send-to-sync)

GCM Flow


Your Server
Google
Device
Server
GCM Server
Device App
Device Receiver
register -> regId
HTTP register (regId)
send (message, devices)
send (message)

GCM in Activity

- onCreate
  - Check if Google Play Services ok
  - Get regId (prefs or GCM)
  - Send regId to our server
  - Save regId in prefs
- onResume
  - Check if Google Play Services ok

GCM Receiving Event

- Broadcast Receiver gets event
- If just a “poke”
  - Notify activity or service
- If needs processing
  - Start service to handle

GCM in Server

- HTTP Post to GCM server
- Header contains
  - Content-type: application/json
  - Authorization:key=xxxxxxxxxx
- Pass JSON containing
  - registration_ids
  - data for message (if more than poke)
- If ids remapped, update registration list

Authorization

- Create application via Google console
- Create server-side key
  - Pass with requests
- When clients register, they pass the project number as a valid sender

Example: Simple GCM

- I've bundled most of the client-side code in a superclass for our Activity…
- Activity registers a broadcast receiver while it's active
- GCM broadcast receiver sends broadcast intent to "poke"
- Activity fetches updates